

# Nuclear Model and Radioactivity



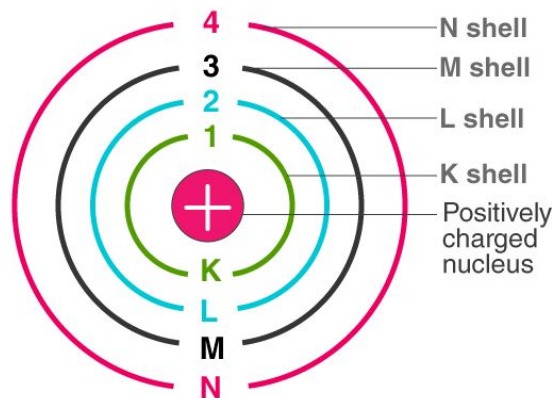
## What is Bohr's Model of an Atom?

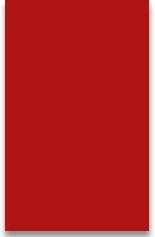
The Bohr model of the atom was proposed by Neil Bohr in 1913. It came into existence with the modification of Rutherford's model of an atom. Rutherford's model introduced the nuclear model of an atom, in which he explained that a nucleus (positively charged) is surrounded by negatively charged electrons.

### Introduction to the Bohr Model

Bohr theory modified the atomic structure model by explaining that electrons move in fixed orbits (shells) and not anywhere in between and he also explained that each orbit (shell) has a fixed energy. Rutherford explained the nucleus of an atom and Bohr modified that model into electrons and their energy levels.

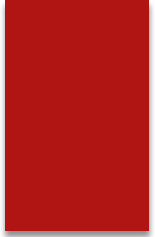
BOHR'S MODEL OF AN ATOM





## Postulates of Bohr's Model of an Atom

- In an atom, electrons (negatively charged) revolve around the positively charged nucleus in a definite circular path called orbits or shells.
- Each orbit or shell has a fixed energy and these circular orbits are known as orbital shells.
- The energy levels are represented by an integer ( $n=1, 2, 3\dots$ ) known as the quantum number. This range of quantum number starts from nucleus side with  $n=1$  having the lowest energy level. The orbits  $n=1, 2, 3, 4\dots$  are assigned as K, L, M, N.... shells and when an electron attains the lowest energy level, it is said to be in the ground state.
- The electrons in an atom move from a lower energy level to a higher energy level by gaining the required energy and an electron moves from a higher energy level to lower energy level by losing energy.



## Limitations of Bohr Atomic Model Theory

- It violates the Heisenberg Uncertainty Principle. The Bohr atomic model theory considers electrons to have both a known radius and orbit i.e. known position and momentum at the same time, which is impossible according to Heisenberg.
- The Bohr atomic model theory made correct predictions for smaller sized atoms like hydrogen, but poor spectral predictions are obtained when larger atoms are considered.
- It failed to explain the Zeeman effect when the spectral line is split into several components in the presence of a magnetic field.
- It failed to explain the Stark effect when the spectral line gets split up into fine lines in the presence of an electric field.

# Atomic Nucleus

## Nucleus

Nucleus of an atom is a tiny positively charged object that contains almost all of its mass. The nucleus is made up of protons and neutrons and contains no electrons.

## Basic properties of Nucleus

- Primary constituents of nuclei are protons and neutrons.
- Nuclear mass is roughly the sum of its constituent protons and neutrons.
- Nuclear charge is +q times the number of protons (Z).

Symbols for Nuclides  ${}^A_ZX$

X is the chemical symbol of the element

Z = Atomic Number = number of proton

N = neutron number = number of neutron

A = Z + N = mass number = Total number of protons and neutrons.

For example arsenic – 75 is represented by  ${}^{75}_{33}\text{As}$

## Atomic mass unit (u)

Atomic mass unit refers to mass of neutral atom not the stripped nuclei. Different atoms of a particular element can have different masses because of different number of protons in their nucleus. These varieties of an element are called isotopes.

Only twenty elements in nature are composed of only one nuclide each. The other elements are composed of two or more isotopes.

Hydrogen has two more isotopes, deuterium and tritium. The atomic masses of the isotopes are 1.007825 u, 2.014102 u, and 3.01605u respectively; however the two heavier isotopes make only 0.015 percent of the natural hydrogen.

The mass of a the most abundant isotope of carbon,  $^{12}_6\text{C}$  is exactly 12 amu.

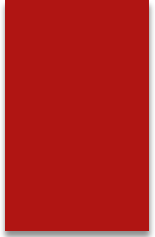
One amu is defined as the 1/12 th mass of  $^{12}_6\text{C}$  atom

The value of 1 amu is  $1 \text{ amu} = 1.66054 \times 10^{-27} \text{ kg}$  whose equivalent energy  
 $= 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2 = 1.49 \times 10^{-10} \text{ J} =$

$$[\text{Mass of } ^{12}_6\text{C} \text{ atom} = 12 \text{ g. Therefore, } 1 \text{ amu} = \frac{1}{12} [\text{mass of } ^{12}_6\text{C} \text{ atom}] = 1 \text{ g.} \quad \frac{1.49 \times 10^{-10}}{1.6 \times 10^{-19}} = 931 \text{ MeV}]$$

Now, one gram atom of carbon contains  $6.02 \times 10^{23}$  atoms.

$$\text{Therefore, } 1 \text{ amu} = \frac{1}{6.022 \times 10^{23}} = 1.66054 \times 10^{-27} \text{ kg}]$$



## Dynamic properties of nuclei

Nuclei can be in an excited state of definite energies. **Transition between excited states occur by emitting  $\gamma$ -rays.** The separation between nuclear states will be  $\sim 10^4$  eV to 1 MeV. From  $\gamma$ -ray spectra, nuclear energy states and nuclear models can be understood.

Nuclei can be transformed into each other. Some of the transformations occur spontaneously by the emission of +ve or -ve electrons ( $\beta$ -rays) or  $\alpha$ -particles. Other transformation can be induced by nuclear bombardments. In these transformations the following quantities must be conserved:

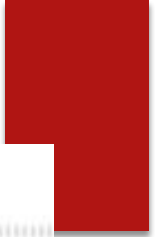
- (i) Number of nucleons
- (ii) Mass
- (iii) Energy
- (iv) Linear momentum
- Angular momentum

## Nuclear Stability

Nuclei of atoms contain protons and neutrons. Positively charged protons repel each other due to electrostatic repulsion between them. This electrostatic repulsion is overcome by the strong nuclear force, the attractive force present between nucleons. Neutrons are important for stabilizing the nucleus. If the attractive force between nucleons is less than the electrostatic repulsion then it makes the nucleus unstable and results in decay.

It defines the stability of an isotope of the elements. Nucleons with high binding energy are more stable. Stability of an isotope can be determined by calculating the ratio of neutrons to protons present in a nucleus ( $N/Z$ ). Elements having atomic number less than 20, mostly have proton and neutron ratio 1:1. The number of neutrons increases as the atomic number increases. Most of the stable nuclei have neutrons to protons ratio more than 1. Only  $^1\text{H}$  and  $^3\text{He}$  have neutrons to protons ratio less than one but are stable.

The first 80 elements of the periodic table have stable isotopes. All the elements with the atomic number more than 82 are unstable and radioactive, irrespective of the number of neutrons.



A 1 to 1 neutron to proton ratio holds true for the stable nuclei of the first twenty elements in the periodic table. This ratio increases to 1.5 to 1 around atomic number 80. Elements above atomic number 83 with 209 nucleons do not exist as stable isotopes. Thus for polonium, with 84 protons, the repulsive forces due to the 84 protons are so large that regardless of the number of neutrons, its nuclides are unstable.

When the neutron to proton ratio is too large or too small, the nucleus is unstable, the atom is called a **radionuclide** and undergoes radioactive decay. If a radionuclide has a



## Nuclear stability and radioactive decay

The neutrons and protons in a nucleus are held together by a combination of forces such as gravitational and electrostatic forces. Some of these forces are repulsive and others are attractive. The protons repel each other. Thus, it becomes difficult for the nucleus to be stable for bigger atoms. The only reason that the nucleus is stable is that the neutrons bind other neutrons or protons. That is why heavier atoms have more neutrons.

In general, for light atoms, there are equal number of neutrons and protons, but for the heavier atoms, there are more neutrons to maintain the stability of the nucleus. Stability is an important factor, because some atoms are unstable.

For example, oxygen has three stable isotopes  ${}^{16}_8\text{O}$ ,  ${}^{17}_8\text{O}$ ,  ${}^{18}_8\text{O}$  [N = 8, 9, 10] and five known unstable (i.e. radioactive) isotopes  ${}^{13}_8\text{O}$ ,  ${}^{14}_8\text{O}$ ,  ${}^{15}_8\text{O}$ ,  ${}^{19}_8\text{O}$  and  ${}^{20}_8\text{O}$  [N = 5, 6, 7, 11, 12]

Uranium  ${}^{235}\text{U}$  Potassium  ${}^{40}\text{K}$  can all be found naturally in the world and they are not stable. Unstable isotopes emit particle or radiation, leaving an isotope which is different from the original one.



## Man-made isotopes

One of the man-made isotopes is cobalt-60 or  $^{60}\text{Co}$ . This  $^{60}\text{Co}$  isotope is used in radiotherapy and is produced by nuclear reaction:



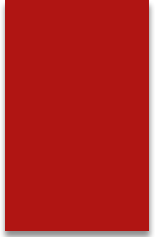
Another example of man-made isotope is technetium-99. This is used for brain scanning. It has 6h half-life. For this short half-life, its mother isotope, which has long half-life, is pressured by special technique. From this mother isotope, the daughter isotope is produced for scanning purpose.

## Radioactivity

Radioactivity is the phenomenon of spontaneous disintegration of heavy atomic nuclei like  $^{238}\text{U}$   $^{235}\text{U}$   $^{232}\text{Th}$   $^{228}\text{Ra}$  etc. through emission of particles or electromagnetic radiations at a constant rate. It is a nuclear property of the active element and in all radioactive processes, a transmutation (change from one element to another) of the elements occurs.

### Characteristics of radioactive substances:

- (i) The radioactive decay occurs naturally, i.e., the radioactive substances emit radiation without any external excitation.
- (ii) Radioactivity is a time dependent property of a nucleus.
- (iii) They continuously go on decreasing in weight.
- (iv) The radioactivity is most found in the substances with high atomic weight.
- (v) Radioactive substances ionize the gas and affect photographic plate.
- (vi) When radioactive element undergoes  $\alpha$  or  $\beta$  decay, its atomic number  $Z$  changes and becomes a nucleus of different element and penetrates through matter.

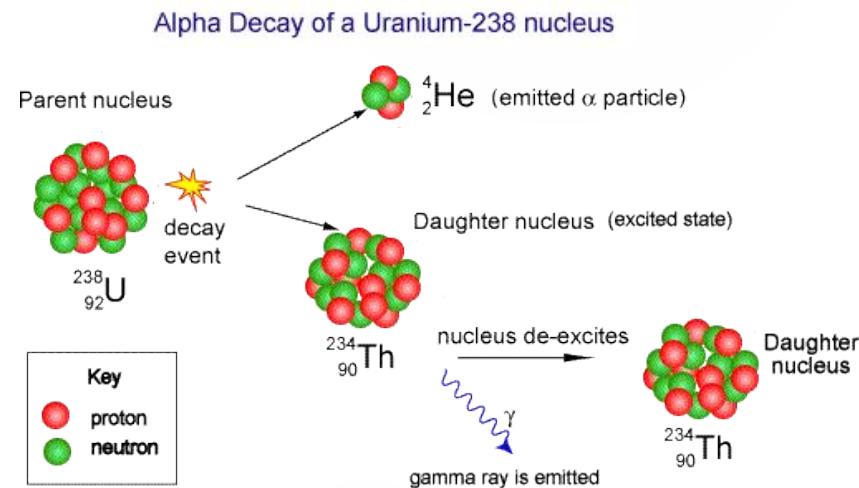
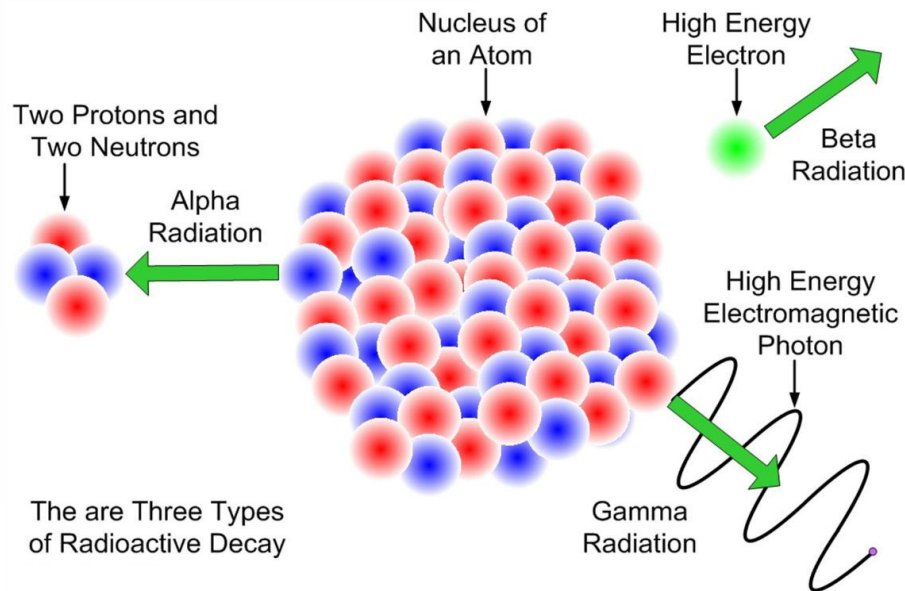


## Radioactivity is a nuclear phenomena

Any physical change (such as change in pressure, temperature) or chemical change (such as excessive heating, freezing, action of strong magnetic and electric fields, chemical treatments, oxidation etc.) does not change the rate of decay and the nature of radiation emitted by the substance. This shows that the phenomena of radioactivity cannot be due to orbital electrons which could be easily affected by such changes. Therefore, the radioactivity should be the property of nucleus. *Thus it can be stated that radioactivity is a nuclear phenomena.*

There are three types of radioactive decays in nature:

- $\alpha$ -decay – a helium nucleus ( ${}^4_2\text{He}$ ) is emitted
- $\beta$ -decay – where electrons or positrons (*particles with the same mass as electrons, but with a charge exactly opposite to that of an electron*) are emitted;
- $\gamma$ -decay – high energy (hundreds of keV or more) photons are emitted.



## $\alpha$ , $\beta$ and $\gamma$ rays

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- **$\alpha$  rays:** The  $\alpha$  rays consist of material particles with a mass 4 times the proton mass and a charge twice that of the proton. They are identified as helium nucleus:  ${}^4_2\text{He}$
- They can ionize gas and penetrate matter and also show scattering (Rutherford  $\alpha$  particle scattering).
- They have the charge  $3.2 \times 10^{-19}\text{C}$

### **$\beta$ rays:**

- The  $\beta$  rays also consist of material particles but with negligible mass and positive charge.
- The  $q/m$  value of  $\beta$  rays agree with that of the electron. Hence  $\beta$  rays are identified as electrons.
- They can ionize gas, affect photographic plate and penetrate matter. Penetrating power is 100 times greater than that of  $\alpha$  rays.

### **$\gamma$ rays:**

- They are non-material and uncharged in character. They identified as electromagnetic radiations of very small wavelength, smaller than X-rays.
- They ionize the gas, affect photographic plate, and penetrate matter.
- Penetrating power is 10-100 times greater than  $\beta$  rays. Velocity is equal to  $c$ .

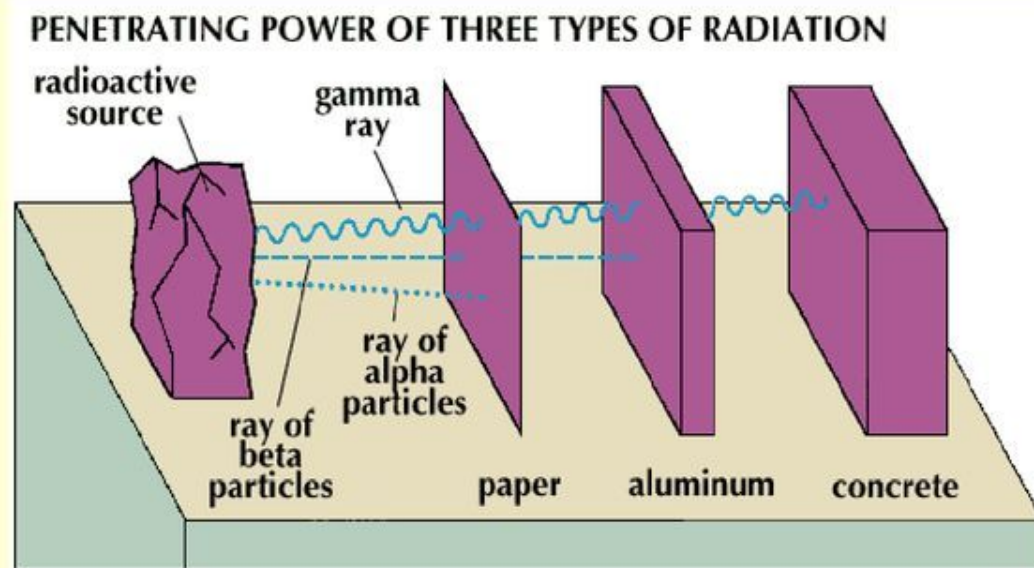
**Table 6.2 Properties of alpha, beta and gamma rays**

| Properties                            | $\alpha$ rays                                                                    | $\beta$ rays                                                                                                                                    | $\gamma$ rays                                                                                                             |
|---------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| What are they?                        | Helium nucleus ( ${}_2\text{He}^4$ ) consisting of two protons and two neutrons. | They are electrons ( ${}_{-1}\text{e}^0$ ), basic elementary particle in all atoms.                                                             | They are electromagnetic waves consisting of photons.                                                                     |
| Charge                                | Positively charged particles. Charge of each alpha particle = $+2e$              | Negatively charged particles. Charge of each beta particle = $-e$                                                                               | Neutral particles. Charge of each gamma particle = zero                                                                   |
| Ionising power                        | 100 time greater than $\beta$ rays and 10,000 times greater than $\gamma$ rays   | Comparatively low                                                                                                                               | Very less ionization power                                                                                                |
| Penetrating power                     | Low penetrating power (even stopped by a thick paper)                            | Penetrating power is greater than that of $\alpha$ rays. They can penetrate through a thin metal foil.                                          | They have a very high penetrating power greater than that of $\beta$ rays. They can penetrate through thick metal blocks. |
| Effect of electric and magnetic field | Deflected by both the fields. (in accordance with Fleming's left hand rule)      | Deflected by both the fields; but the direction of deflection is opposite to that for alpha rays. (in accordance with Fleming's left hand rule) | They are not deflected by both the fields.                                                                                |
| Speed                                 | Their speed ranges from 1/10 to 1/20 times the speed of light.                   | Their speed can go up to 9/10 times the speed of light.                                                                                         | They travel with the speed of light.                                                                                      |

# Penetrating Power of Radiation

- The penetrating power of alpha rays, beta rays, and gamma rays varies greatly.

- Alpha particles can be blocked by a few pieces of paper.
- Beta particles pass through paper but are stopped by aluminum foil.
- Gamma rays are the most difficult to stop and require concrete, lead, or other heavy shielding to block them.



## Alpha Decay

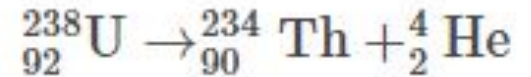
Alpha decay is a type of radioactive decay where the unstable atomic nuclei emit a helium nucleus (alpha particle) and in the process transforms into another more stable element.

The particle ejected out, the alpha particle, consists of four nucleons, and they are two neutrons and two protons.

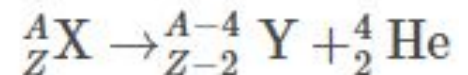
Alpha radiation reduces the ratio of protons to neutrons in the parent nucleus, bringing it to a more stable configuration.

An alpha particle is identical to the nucleus of a helium atom. As the alpha particle; made of two protons and neutron; exits the nucleus, the atomic number of the radioactive sample changes. The element left behind after alpha decay is two atomic numbers lesser and four mass numbers lower.

For example, Uranium  $^{238}_{92}\text{U}$  decays to form Thorium  $^{234}_{90}\text{Th}$ .



Thus, the transformation of a nucleus in an alpha decay can be written as;



## Beta Decay

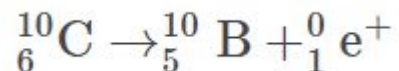
Beta Decay is a type of radioactive decay in which a proton is transformed into a neutron or vice versa inside the nucleus of the radioactive sample.

Processes like beta decay allow the nucleus of the radioactive sample to get as close as possible to the optimum neutron/ proton ratio. While doing so, the nucleus emits a beta particle which can either be an electron or positron. Remember that there either a proton can turn into a neutron or a neutron into a proton.

### What happens in Beta Decay?

Beta decay is a radioactive decay in which a beta ray is emitted from an atomic nucleus. During beta decay, the proton in the nucleus is transformed into a neutron and vice versa. If a proton is converted to a neutron, it is known as  $\beta^+$  decay. Similarly, if a neutron is converted to a proton, it is known as  $\beta^-$  decay. Due to the change in the nucleus, a beta particle is emitted. The beta particle is a high-speed electron when it is a  $\beta^-$  decay and a positron when it is a  $\beta^+$  decay. Beta particles are used to treat health conditions such as eye and bone cancer and are also used as tracers.

the  $\beta^+$  decay of carbon-10 can be represented by an equation as follows:



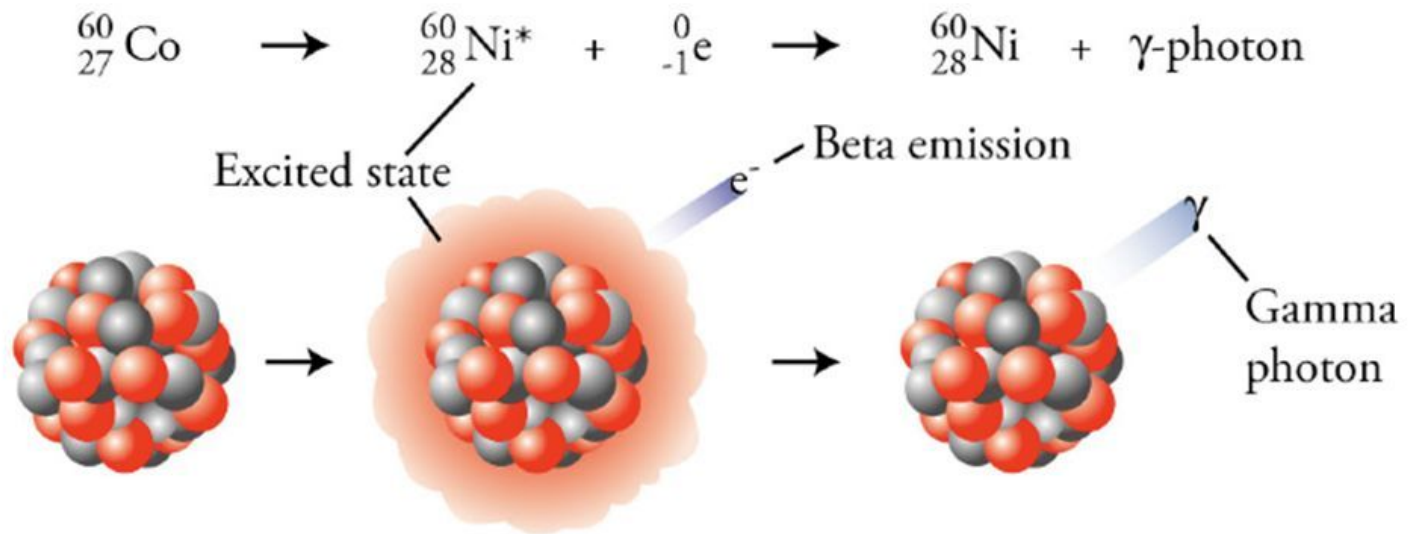
Here, the proton of the carbon atom is converted into a neutron, and the emitted beta particle is a positron.

## What is Gamma Decay?

Gamma decay is the emission of electromagnetic radiation of an extremely high frequency i.e. very high energy, giving out excess energy in order to stabilize the **unstable nucleus**. You must be quite familiar with the various energy levels in an atom. The Nucleus has its own energy levels. Gamma decay is the nucleus's way of dropping from a higher energy level to a lower energy level through the emission of high energy photons. The energy level transition energies in the atom are in the order of MeV. Therefore, the gamma-ray emitted is also of very high energy of the order of MeV, just like x-rays. The gamma rays emitted can be differentiated from x-rays only by the fact that gamma rays come from the nucleus. Due to their high energy, they are extremely penetrating and thereby dangerous to biological life forms.

- the parent nucleus does not undergo any physical change in the process, daughter and parent nuclei are the same.
- Most of the time, gamma decay occurs after the radioactive nuclei have undergone an alpha or a beta decay.
- The alpha and beta decays leave the daughter nuclei in an excited state. From the excited state, the daughter nuclei can get back to the ground state by emitting one or more high energy gamma ray

# Gamma Emission Example:



## Activity

The activity of a sample of radioactive nuclide is the rate at which the nuclei of its constituent atoms decay. If  $N(t)$  is the number of radioactive nuclei present in a sample at a certain time its activity  $A$  is given by  $-\frac{dN(t)}{dt}$

The S.I. unit of activity is named after Henry Becquerel: 1 becquerel = 1 Bq = 1 events/s

The activities encountered in practice are some times so high that the mega becquerel ( 1 MBq =  $10^6$  Bq) and giga becquerel (1 GBq =  $10^9$  Bq) are more appropriate. The traditional unit of activity is curie(Ci) which was originally defined as the activity of 1 g of radium. It is now defined as

$$1 \text{ curie} = 1 \text{ Ci} = 3.7 \times 10^{10} \text{ events /sec} = 37 \text{ GBq}$$

## Laws of radioactive decay

Radioactivity is a physical phenomenon which occurs when an unstable nucleus of an atom transforms itself through a radioactive decay, reaching a new condition of stability. The rate of decay, i.e. the activity of a sample at an instant of time is directly proportional to the number of radionuclide present at that instant.

$$-\frac{dN(t)}{dt} \propto N(t)$$

from which the following equation can be written  $-\frac{dN(t)}{dt} = \lambda N(t)$

here the constant  $\lambda$  is called the “decay constant”.

Let us suppose that at the beginning of disintegrations i.e. at  $t=0$ , the number of radioactive nuclei present in the sample is  $N_0$ .

$$\frac{dN}{N} = -\lambda dt \Rightarrow \int_{N_0}^N \frac{dN}{N} = -\lambda \int_0^t dt' \Rightarrow \ln [N / N_0] = -\lambda t$$
$$\therefore N(t) = N_0 e^{-\lambda t} \quad \text{.....} \quad \text{.....} \quad \text{.....} \quad (3)$$

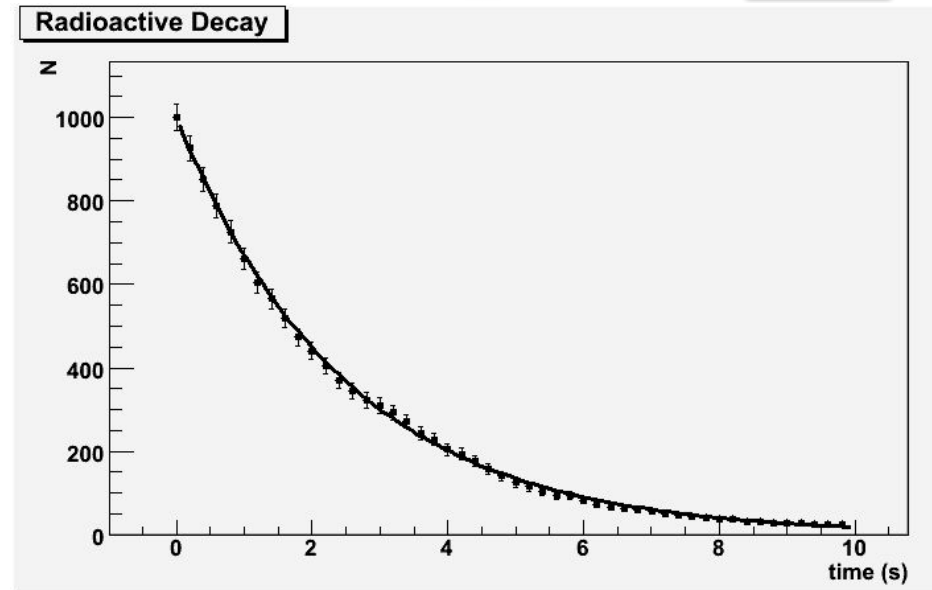
Eq (3) shows that the number of surviving nuclei at any time  $t$  decreases exponentially with time.

From [3], we can write

$$\lambda N(t) = \lambda N_0 e^{-\lambda t}$$

$$\text{or } A(t) = A_0 e^{-\lambda t} \quad [4]$$

where  $A(t)$  is the activity at time  $t$  and  $A_0$  is the activity at time  $t = 0$ .



## Half life:

This concept is widely used in nuclear physics and explains how quickly atoms undergo radioactive decay. In addition, it also means how long an atom will survive radioactive decay.

It is defined as the time interval during which one half of the total number of nuclei that were present at the beginning of the time interval have decayed.

If  $N_1$  nuclei are present at time  $t_1$   
and one half that number

$N_2 = N_1 / 2$  have survived at time  $t_2$ ,

we can write

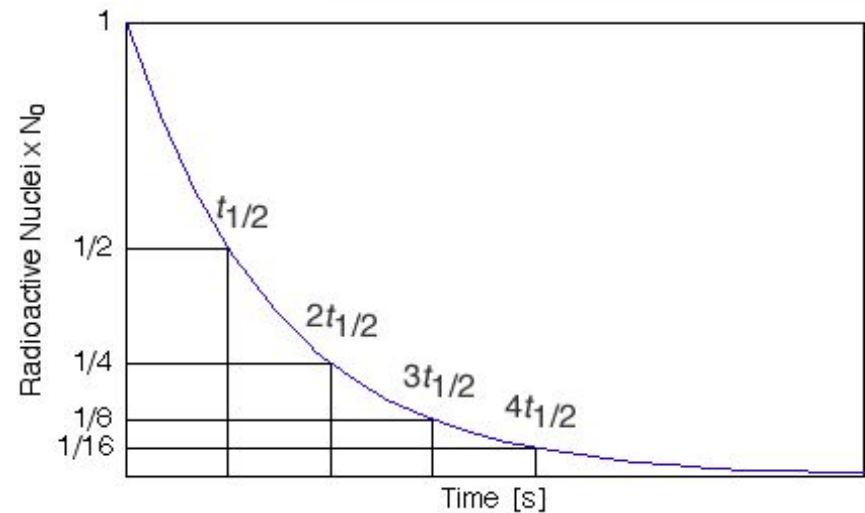
$$N_1 = N_0 e^{-\lambda t_1}$$

$$N_2 = N_0 e^{-\lambda t_2}$$

$$\therefore \frac{N_1}{N_2} = e^{\lambda[t_2 - t_1]} = 2$$

$$\therefore \lambda[t_2 - t_1] = \ln 2$$

$$\therefore T_{1/2} = \text{half life} = [t_2 - t_1] = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$



## Average or mean lifetime:

In a radioactive substance, some of the nuclei decay earlier than others which survive longer. We do not know why this happens, but we know that this decay is a probabilistic phenomenon. Thus an average life time

$\bar{T}$  is introduced to know the average life expectancy of all the nuclei present at any instant before disintegration in a decay process.

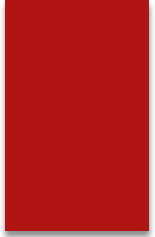
Thus the average life time ( $\bar{T}$ ) of a radioactive element is defined as the ratio of the total life time of all the atoms to the total number of atoms.

$$\bar{T} = \frac{\text{total life time}}{\text{total number of atoms}} = \frac{t_1 dN_1 + t_2 dN_2 + t_3 dN_3 + t_4 dN_4 + \dots}{dN_1 + dN_2 + dN_3 + dN_4 + \dots} = \frac{\sum t dN}{\sum dN}$$

If we introduce the integral instead of summation,

$$\bar{T} = \frac{\int_0^{\infty} t |dN|}{N_0} = \frac{\int_0^{\infty} t \lambda N_0 e^{-\lambda t} dt}{N_0}$$

$$\begin{aligned}
\overline{T} &= \lambda \int_0^{\infty} t e^{-\lambda t} dt \\
&= \lambda \left[ t \int_0^{\infty} e^{-\lambda t} dt - \int_0^{\infty} \left\{ \frac{d(t)}{dt} \int_0^{\infty} e^{-\lambda t} dt \right\} dt \right] \\
&= \lambda \left[ t \int_0^{\infty} e^{-\lambda t} dt - \int_0^{\infty} \frac{e^{-\lambda t}}{-\lambda} dt \right] \\
&= \lambda \left[ \frac{t e^{-\lambda t}}{-\lambda} - \frac{e^{-\lambda t}}{\lambda^2} \right]_0^{\infty} \\
&= - \left[ t e^{-\lambda t} + \frac{e^{-\lambda t}}{\lambda} \right]_0^{\infty} \\
&= - \left[ (0 + 0) - \left( 0 + \frac{1}{\lambda} \right) \right] \\
&= \frac{1}{\lambda}
\end{aligned}$$



## Significance

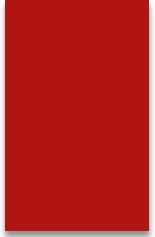
Mean-life plays important role in the area of radioactive isotope; these are mentioned below;

- 1.It helps in identifying the exact dates of artifacts (in calculating the age of any substance).
- 2.It helps in the calculation of time for storing any radioactive waste until they become safe.
- 3.It also supports the doctors in analysing and use of safe radioactive tracers.
- 4.It is a very useful method to assess the rate of the decay process.

## Uses of Radioactivity

Some radioactivity uses are provided in the points below.

- Americium-241 is an alpha emitter and is used for domestic smoke detectors in the United States.
- The alpha particles given out by the Americium sample ionize the air in the chamber of the smoke detector leading to a small current in the chamber.
- When smoke enters the chamber, it causes a drop in current causing the alarm to go off. Although Alpha particles have a very short range, they are devastating when in close contact.
- Alpha emitters, when swallowed, come in close contact with tissue and are deadly in such circumstances and is therefore used in assassination attempts by radiation poisoning.



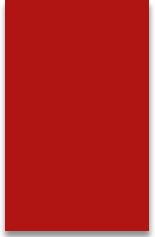
## Advantages and Disadvantages of Radioactivity

### **Advantages of radioactivity are:**

- Gamma rays are used to kill cancerous cells and hence used in radiotherapy.
- Cobalt-60 is used to destroy carcinogenic cells.
- Gamma rays are used in scanning the internal parts of the body.
- Gamma rays kill microbes present in food and prevent it from decay by increasing the shelf life.
- Age of the rocks can be studied using radioactive radiations by measuring the argon content present in the rock.

### **Disadvantages of radioactivity are:**

- High dosage of radioactive radiation on the body might lead to death.
- Radioactive isotopes are expensive.

- 
1. Half life of a radio active substance is 30 days. Calculate the radioactive decay constant, the mean life, the time taken for  $\frac{3}{4}$  th of the original number of atoms to disintegrate and the time for  $\frac{1}{8}$  th of the original number of atoms to remain unchanged.
  2. The half life of radium is 1620 years. In how many years will one gram of pure element lose one centigram and be reduced to one centigram?
  3. 1 gram of radium is reduced by 2.1 mg in 5 years by alpha decay. Calculate the half life.